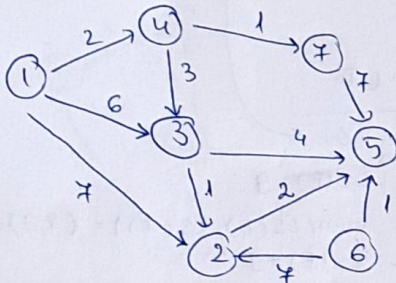


Jun 2023, 1.

1. Digraph, min. cost paths from ① to each vertex, using Dijkstra. (prev)



x	2	d						next prev							
	1	1	2	3	4	5	6	7	1	2	3	4	5	6	7
1	4, 3, 2	0	7	6	2	∞	∞	∞	-	1	1	1	-	-	-
4	7, 3, 2	0	7	5	2	∞	∞	3	-	1	4	1	-	-	4
7	3, 2, 5	0	7	5	2	10	∞	3	-	1	4	1	7	-	4
3	2, 5, 1	0	6	5	2	10	∞	3	-	3	4	1	7	-	4
2	5	0	6	5	2	8	∞	3	-	3	4	1	2	-	4
5	-	0	6	5	2	8	∞	3	-	3	4	1	2	-	4

1: 1, cost 0

2: 1 → 4 → 3 → 2, cost 6

3: 1 → 4 → 3, cost 5

4: 1 → 4, cost 2

5: 1 → 4 → 3 → 2 → 5, cost 9

6: unreachable, cost ∞

7: 1 → 4 → 7, cost 3

2. Activities; earliest, latest schedulings

- critical activities (topol. sorting)
- starting & ending times.

(*) Earliest start (ES) = maximum earliest finish of all preneg.

Earliest finish (EF) = ES + dur.

(*) Latest finish (LF) = min. latest start of all successors

Latest start (LS) = LF - dur

Activity	Duration	Prerequisites
→ A	3	C
→ B	4	C, D
→ C	3	-
→ D	3	-
→ E	1	A, B, F
→ F	2	B

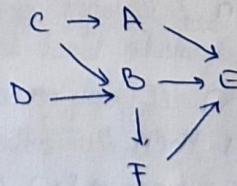
C → A

C → B; D → B

A → E; B → E; F → E

B → F

⇒ Activities as graph:



⇒ Topological order: C, D, A, B, F, E.

ES	Activity	Duration	ES	EF
(*)	C	3	0	0 + duration = 0 + 3 = 3.
start from the activity having the least preneg.	D	3	0	0 + 3 = 3.
	A	3	EF(C) = 3	3 + 3 = 6.
	B	4	max(EF(C), EF(D)) = 3	3 + 4 = 7.
	F	2	max(6, 7) = 7	7 + 2 = 9.

E

1

$$\max(6, 7, 9) = 9$$

$$9 + 1 = 10$$

proj. duration.

(4)

LS:

Activity

Duration

LS

LF

start from
the activity
having the
most presig.

E

1

$$10 - 1 = 9$$

10

F

2

$$9 - 2 = 7$$

$$LS(E) = 9$$

B

4

$$7 - 4 = 3$$

$$\min(LS(E), LS(F)) = (9, 7) = 7$$

A

3

$$9 - 3 = 6$$

$$LS(E) = 9$$

C

3

$$3 - 3 = 0$$

$$\min(LS(A), LS(B)) = (6, 3) = 3$$

D

3

$$3 - 3 = 0$$

$$LS(B) = 3$$

3) Polynomial time alg. $\rightarrow$ directed acyclic graph (DAG)
 $\rightarrow$ nr. of $\neq$ paths of max. length between 2 vertices.

Input: G, s, t

~~topo = topological sort~~ (G)

$x_0, \dots, x_{m-1} = \text{toposort}(G)$

for x in X do
 $d[x] = -\infty$
 $\text{count}[x] = 0$

no vertex accessible
from (s) .

$d[s] = 0$
 $\text{count}[s] = 1$

one path from (s) to (s)
 $\Rightarrow$ length 0.

for $i = 0, m-1$ do
 $x = x_i$

process in topological order

for y in $\text{Nout}(x)$ do // $x \rightarrow y$

if $d[y] < d[x] + 1$ then // longer path to y
 $d[y] = d[x] + 1$ // update best length

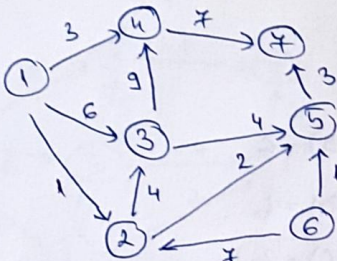
$\text{count}[y] = \text{count}[x]$ // copy nr. of best paths to x .

else if $d[y] = d[x] + 1$ then // another path to y with the same length
 $\text{count}[y] = \text{count}[y] + \text{count}[x]$ // copy these paths too.

return $\text{count}[t]$ // from (s) to (t) .

Jun 2023, 2.

1. Digraph, min. cost paths from ① to each vertex, using Dijkstra.



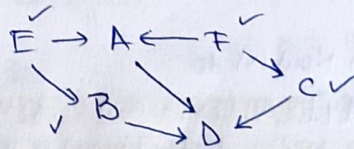
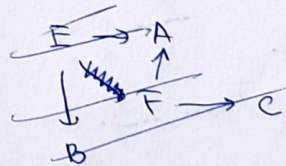
g = where can I go from x

x	g	d	prev
	1	1 2 3 4 5 6 7 ∞ ∞ ∞ ∞ ∞ ∞ ∞	1 2 3 4 5 6 7 - - - - - - -
1	2, 4, 3	0 1 6 3 ∞ ∞ ∞	- 1 1 1 - - -
2	4, 5, 3	0 1 5 3 ∞ ∞	- 1 2 1 2 - -
4	5, 3, 7	0 1 5 3 3 ∞ 6	- 1 2 1 2 - 4
5	3, 7	0 1 5 3 3 ∞ 6	- 1 2 1 2 - 5
3	7	0 1 5 3 3 ∞ 6	- 1 2 1 2 - 5
7		0 1 5 3 3 ∞ 6	- 1 2 1 2 - 5

- 1: 1, cost 0
- 2: 1→2, cost 1
- 3: 1→2→3, cost 5
- 4: 1→2→3→1→4, cost 3
- 5: 1→2→5, cost 3
- 6: unreachable, cost ∞
- 7: 1→2→5→7, cost 6.

2) Earliest, latest schedulings
Critical activities (topol. sorting)
Starting, ending times.

Activ.	Duration	Prereg.
A	2	E, F ✓
B	1	E ✓
C	3	F ✓
D	5	A, B, C
E	4	-
F	3	-



⇒ Topological order: E F C B A D

ET: $ES(\text{node}) = \max$ earliest finish of all prereq.
 $EF(\text{node}) = ES + \text{duration}$.

Activ	Duration	ES	EF
E	4	0	0+4=4
F	3	0	0+3=3
C	3	$EF(F)=3$	3+3=6
B	1	$EF(E)=4$	4+1=5
A	2	$\max(3, 4)=4$	4+2=6
D	5	$\max(6, 5, 6)=6$	6+5=11

⇒ proj. duration = 11.

LT: $LF(\text{node}) = \min.$ latest start of all successors.

$LS(\text{node}) = LF - \text{duration}.$

Activity	Duration	LS	LF
D	5	$11 - 5 = 6$	11
A	2	$6 - 2 = 4$	6
B	1	$6 - 1 = 5$	$LS(D) = 6$
C	3	$6 - 3 = 3$	$LS(D) = 6$
F	3	$3 - 3 = 0$	$LS(C) = 3$
E	4	$4 - 4 = 0$	$\min(LS(A), LS(B)) = \min(4, 5) = 4$

3) Polynomial time alg. $\rightarrow$ DAG w costs and 2 vertices
 $\rightarrow$ nr. of $\neq$ paths of **min. cost** between 2 vertices.

Input: G, s, t ; $G(X, E)$; $X = \text{set of vertices}$, $E = \text{set of edges}$.

Output: nr. of min-cost paths from s to t .

$x_0, \dots, x_{m-1} = \text{toposort}(G)$

for x in X do

$\text{dist}[x] = \infty$

$\text{count}[x] = 0$

$\text{dist}[s] = 0$

$\text{count}[s] = 1$

for $i = 0, m-1$ do

$x = x_i$

 for y in $\text{Nout}(x)$ do

 if $\text{dist}[y] > \text{dist}[x] + \text{cost}(x, y)$ then

$\text{dist}[y] = \text{dist}[x] + \text{cost}(x, y)$

$\text{count}[y] = \text{count}[x]$

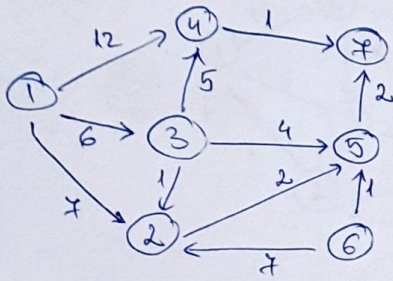
 else if $\text{dist}[y] == \text{dist}[x] + \text{cost}(x, y)$ then

$\text{count}[y] = \text{count}[y] + \text{count}[x]$

return $\text{count}[t]$

Jan 2023, 3

1. Digraph, min. cost path to 7, reversed Dijkstra.



2 = what goes in x.

x	2	d	next
	7	1 2 3 4 5 6 7 ∞ ∞ ∞ ∞ ∞ ∞ 0	1 2 3 4 5 6 7 - - - - - - -
7	4, 5	∞ ∞ ∞ 1 2 ∞ 0	- - - 7 7 - -
4	1, 3, 5	13 ∞ 6 1 2 ∞ 0	4 - 4 7 7 - -
5	2, 6, 1, 3	13 4 6 1 2 3 0	4 5 4 7 7 5 -
6	2, 1, 3	13 4 6 1 2 3 0	4 5 4 7 7 5 -
2	1, 3	11 4 5 1 2 3 0	2 5 2 7 7 5 -
3	1	11 4 5 1 2 3 0	2 5 2 7 7 5 -
	-	11 4 5 1 2 3 0	2 5 2 7 7 5 -

$$d[2] + 7 = 4 + 7 = 11$$

$$d[2] + 1 = 5$$

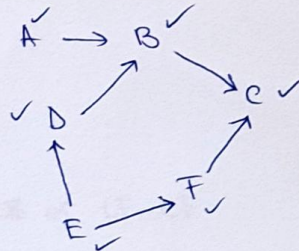
$$d[3] + 1 = 6 + 1 = 7$$

x=2: 1: $d[2] + (1,2) = 4 + 7 = 11 < 13 \Rightarrow \text{upd}$
3: $d[2] + (3,2) = 4 + 1 = 5 < 6 \Rightarrow \text{upd}$

x=3: 1: $d[3] + (1,3) = 5 + 6 = 11 > 5$

2) ET, LT, critical activities;

Act.	Duration	Prereq.
A	5	-
B	4	A, D
C	1	B, F
D	3	E
E	2	-
F	5	E



AEDBC

topological: AEDBDFC
AEDFBC

ET:

Activ.	Duration	ES	EF
A	5	0	5
E	2	0	2
D	3	EF(E)=2	2+3=5
F	5	EF(E)=2	2+5=7
B	4	max(5,2)=5	5+4=9
C	1	max(7,9)=9	9+1=10

proj. duration = 10

LT:

Activ	Duration	LS	LF
C	1	10-1=9	10
B	4	9-4=5	LS(C)=9
F	5	9-5=4	LS(C)=9
D	3	5-3=2	LS(B)=5
E	2	2-2=0	min(4,2)=2 LS(F)=4
A	5	5-5=0	LS(B)=5

what enters in D, F, B, C.

what comes out of B, F, D, E, A.

Activity	<u>ES</u>	<u>EF</u>	<u>LS</u>	<u>LF</u>	Slack
A	0	5	<u>0</u>	5	$ES-LS=0$; $EF-LF=0$.
E	0	2	<u>0</u>	2	✓
D	2	5	<u>2</u>	5	✓
F	<u>2</u>	7	<u>4</u>	<u>9</u>	7 X
B	<u>5</u>	9	<u>5</u>	<u>9</u>	✓
C	9	10	<u>9</u>	<u>10</u>	✓

⇒ critical activ: A, E, D, B, C.

critical paths: $A \rightarrow B \rightarrow C \rightarrow \text{cost: } 5 + 4 + 1 = 10$

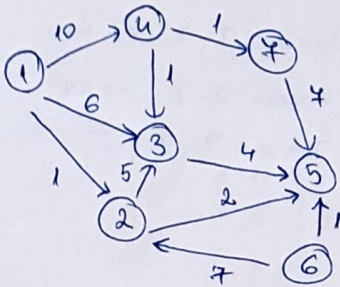
$E \rightarrow D \rightarrow B \rightarrow C \rightarrow \text{cost: } 2 + 3 + 4 + 1 = 10$

⇒ They are both critical.

3) see Jun 2023, 2.

Jun 2023, 4

1. Digraph, min cost path to vertex 7, reversed Dijkstra.



z = where can I go from
 g = what goes in x .

x	z	d	next
	7	0 2 3 4 5 6 7 $\infty \infty \infty \infty \infty \infty 0$	1 2 3 4 5 6 7 - - - - - - -
7	4	$\infty \infty \infty 1 \infty \infty 0$	- - - 7 - - 4
4	3	11 $\infty \infty 1 \infty \infty 0$	4 - - 7 - - -
1	-	11 $\infty \infty 1 \infty \infty 0$	4 - - 7 - - -

1: $1 \rightarrow 4 \rightarrow 7$, cost 11

2: ∞

3: ∞

4: $4 \rightarrow 7$, cost 10

5: ∞

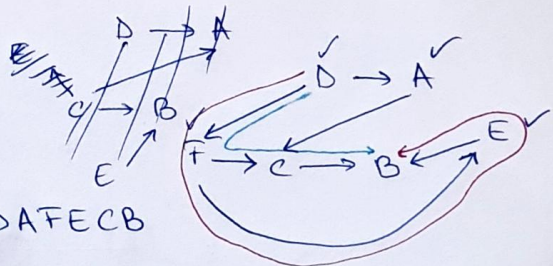
6: ∞

7: 7, cost 0.

2.

Activ.	Duration	Prereg.
A	1	D
B	2	C, E
C	3	A, F
D	1	-
E	3	F
F	3	D

LS-ES or LF-EF
 "
 if slack = 0 $\Rightarrow$ critical activities.



topo: D A F E C B

ET: "what enters in Activ."

Activ.	Duration	ES	EF
D	1	0	0+1=1
A	1	EF(D)=1	1+1=2
F	3	EF(D)=1	1+3=4
E	3	EF(F)=4	4+3=7
C	3	EF(E)=4 ^{max} = 4	4+3=7
B	2	max = 7	7+2=9

LT:

Activ.	Duration	LS	LF
B	2	9-2=7	9
C	3	7-3=4	LS(B)=7
E	3	7-3=4	LS(B)=7
F	3	4-3=1	ES min(4,4)=1
A	1	4-1=3	LS(C)=4
D	1	1-1=0	min(1,3)=1

Activity	$\overline{ES}$	$\overline{EF}$	$\overline{LS}$	$\overline{LF}$	slack
D	0	1	0	1	$0-0=0=1-1 \Rightarrow$ critical activ.
A	1	2	3	4	$3-1=2=4-2 \neq 0$
F	1	4	1	4	$1-1=0=4-4 \Rightarrow$ critical activ.
E	4	7	4	7	$4-4=0=7-7 \Rightarrow$ cr.
C	4	7	4	7	$4-4=0=7-7 \Rightarrow$ cr.
B	7	9	7	9	$7-7=0=9-9 \Rightarrow$ cr.

critical activities: D, F, E, C, B.

critical path : $D \rightarrow F \rightarrow E \rightarrow B$: duration: $1+3+3+2=9$
 $D \rightarrow F \rightarrow C \rightarrow B$: duration: $1+3+3+2=9$ } $\Rightarrow$

$\Rightarrow$ they are both critical paths.

! The critical path is the longest path if there are more.

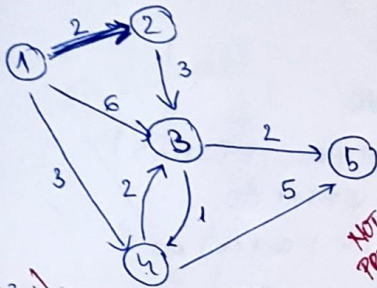
3) Polyn. time alg $\rightarrow$ DAG, 2 vertices

$\rightarrow$ nr. of paths of max length between 2 vertices.

(see Jun 2023, 1).

Jun 2018, 2.

1) Digraph, min. cost path from ① to each vertex; Dijkstra (prev).



NOT FROM THE PREVIOUS!

x	2	d	prev
	1	1 2 3 4 5	1 2 3 4 5
1	2, 3, 4	0 2 6 3 ∞	- 1 1 1 5
2	3, 4	0 2 5 3 ∞	- 1 2 1 -
3	5	0 2 5 3 8	- 1 2 1 4
4	5	0 2 5 3 7	- 1 2 1 3

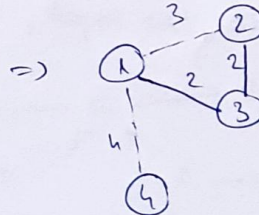
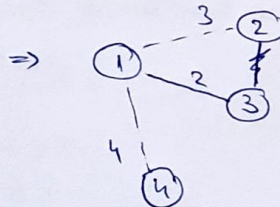
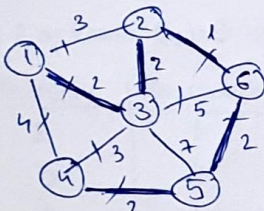
$x=3$:
 $d=4$: newcost = $d[3] + 1 = 5 + 1 = 6$
 $d[4] = 3$
 $\Rightarrow$ no update.
 $d=5$: newcost = $d[3] + 2 = 5 + 2 = 7$
 $d[5] = \infty$
 $\Rightarrow$ update.
 $x=4$:
 $d=5$: newcost = $d[4] + 1 = 3 + 1 = 4$
 $d[5] = 7$
 $\Rightarrow$ update.

2) min. spanning tree, Prim's.

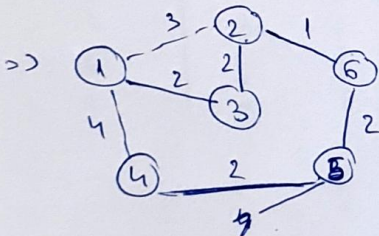
$\rightarrow$ "expand current tree".

$\rightarrow$ look only at edges touching the tree.

$\rightarrow$ don't sort edges.



$\Rightarrow$



min. sp. tree, Kruskal's.

$\rightarrow$ sort edges.

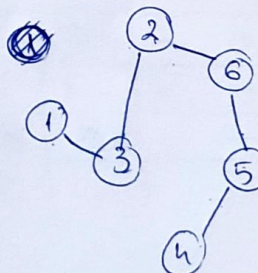
~~4-5 (3)~~

Vertices: 6

$\downarrow$

Stop when $6 - 1 = 5$ edges.

2-6: 1
 2-3: 2
 1-3: 2
 6-5: 2
 5-4: 2
 3-4: 3
 1-2: 3
 4-1: 4
 3-6: 5



$n - 1 = 6 - 1 = 5$ edges.
 $\Rightarrow$ STOP.

3) DG, cycle of min. length, if $\exists$.

(BFS) helper:

Queue q

Dictionary prev, dist

~~for x~~

Set visited

q.enqueue(s)

visited.add(s)

dist[s] = 0

prev[s] = Nil

while not q.isEmpty() do

 x = q.dequeue()

 for y in Nout(x) do

 if y not in visited then

 visited.add(y)

 dist[y] = dist[x] + 1

 prev[y] = x

 q.enqueue(y)

return visited, dist, prev.

bestCycle = Nil

bestLength = ∞

for s in X do

 visited, dist, prev = BFS(G, s)

 for x in visited do

 for y in Nout(x) do

 if y == s then

 cycleLen = dist[x] + 1

 if cycleLen < bestLen then

 bestLen = cycleLen

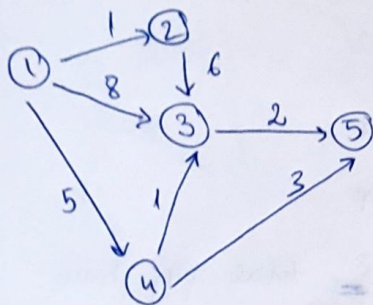
 bestCycle = s $\rightarrow$ x $\rightarrow$ x $\rightarrow$ s

return bestCycle.

Jun 2018, 5

next.

1) Graph, min cost path from all vertices to ⑤, reversed Dijkstra.



$x=3$;

$d=1$: newcost = $d[3] + 1 =$

$d=4$: newcost = $d[3] + 1 = 2 + 1 = 3$
 $d[4] = 3$

$x=4$

$d=1$: newcost: $d[4] + 5 = 3 + 5 = 8$.
 $d[1] = 10$
 = update.

$d=2$: newcost = $d[1] +$

$x=2$

$d=$

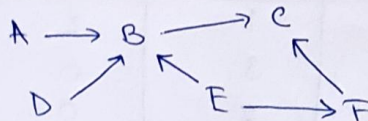
x	2	d	next
	5	1 2 3 4 5 $\infty \infty \infty \infty 0$	1 2 3 4 5 - - - - -
5	3, 4	$\infty \infty 2 3 0$	- - 5 5 -
3	4, 1, 2, 3	10 8 2 3 0	3 3 5 5 -
4	1, 2	8 8 2, 3, 0	4 3 5 5 -
2	1	8 8 2 3 0	4 3 5 5 -
1	-		

final paths to 5:

- 1: 1 → 4 → 5, cost 8
- 2: 2 → 3 → 5, cost 8
- 3: 3 → 5, cost 2
- 4: 4 → 5, cost 3
- 5: 5, cost 0.

2) Earliest, latest scheduling, critical activ., topol. sort, ST, ET.

Activity	Duration	Prereq.
A	1	-
B	4	A, D, E
C	1	B, F
D	3	-
E	2	-
F	5	E



topological sort: A D E F B C

x	2	inbound
		A B C D E F
	A, D, E	0 3 2 0 0 1
A	D, E	- 2 2 0 0 1
D	E	- 1 2 - 0 1
E	F, B	- 0 2 - - 0
F	B, C	- - 1 - - -
B	C	- - 0 - - -
C		

current
indegree = 0

current
indegree = 0

choice matters
(no indegree = 0)

Activity	Duration	Pre-req. ES	EF ↓	"what enters in activ."
A	1	0	$0+1=1$	<u>MAX</u>
D	3	0	$0+3=3$	
E	2	0	$0+2=2$	
F	5	$EF(E)=2$	$2+5=7$	
B	4	$\max(1,3,2)=3$	$3+4=7$	
C	1	$\max(7,7)=7$	$7+1=8$	total proj. time.

Activ.	Duration	LS ↓	LF	"what leaves the activ."
C	1	$8-1=7$	8	Min.
B	4	$7-4=3$	$LS(C)=7$	
F	5	$7-5=2$	$LS(C)=7$	
E	2	$2-2=0$	$\min(LS(B), LS(F))=2$	
D	3	$3-3=0$	$\min(LS(E), LS(F))=3$	
A	1	$3-1=2$	$LS(B)=3$	

Activ.	ES	EF	LS	LF
A	0	1	2	3
D	0	3	0	3
E	0	2	0	2
F	2	7	2	7
B	3	7	3	7
C	7	8	7	8

2) D, E, F, B, C are critical

paths: $D \rightarrow B \rightarrow C$: ~~cost~~ duration: $3+4+1=8$.

$E \rightarrow F \rightarrow C$: duration: $2+5+1=8$. = proj. duration

$E \rightarrow B \rightarrow C$: duration: $2+4+1=7$. $\neq$ proj. duration

=> critical paths: $D \rightarrow B \rightarrow C$

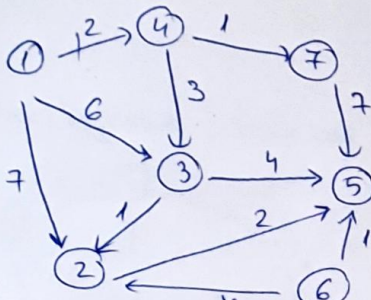
$E \rightarrow F \rightarrow C$

3) ~~see~~

3) neg. cycles -> see ref. sheet.

Jun 2024, 1

1) Dijkstra; from 1 to all;



$x=4$

$$d=3: d[4]+3=2+3=5, \text{ ✗}$$

$$d[3]=6$$

$$d=7: d[4]+1=3 \checkmark$$

d

$$x=7: d=5; d[7]+7=10$$

$$d[5]=\infty$$

$$x=3; d=2; d[3]+1=6 < d[2]=7$$

x	g	d	pref.
		1 2 3 4 5 6 7	1 2 3 4 5 6 7
		0 ∞ ∞ ∞ ∞ ∞ ∞	← - - - - -
1	2, 3, 4	0 7 6 2 ∞ ∞ ∞	1 1 1 - - -
4	2, 3, 7	0 7 5 2 ∞ ∞ 3	- 1 4 1 - - 4
7	2, 3, 5	0 7 5 2 10 ∞ 3	- 1 4 1 7 - 4
3	2, 5	0 6 5 2 9 ∞ 3	- 3 4 1 3 - 4
2	5	0 6 5 2 8 ∞ 3	- 3 4 1 2 - 4

$x=8$

$$d=5; d[2]+2=8$$

$$d[5]=9$$

$$\checkmark 1: 1, \text{ cost } 0$$

$$\checkmark 2: 1 \rightarrow 4 \rightarrow 3 \rightarrow 2, \text{ cost } 6$$

$$\checkmark 3: 1 \rightarrow 4 \rightarrow 3, \text{ cost } 5$$

$$\checkmark 4: 1 \rightarrow 4, \text{ cost } 2$$

$$\checkmark 5: 1 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 5, \text{ cost } 8$$

$$\checkmark 6: \infty$$

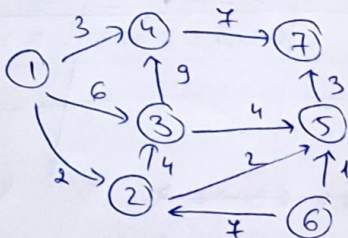
$$\checkmark 7: 1 \rightarrow 4 \rightarrow 7, \text{ cost } 3.$$

2) see 14) from "Patriuanciu exam prep".

3) see ref. sheet (DAG, w.r. paths, max. cost.)

Jun 2024, 2

1) Dijkstra, from 1 to all;

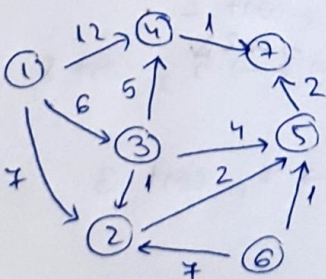


2) see 16) from "Patriuanciu..."

3) see ref. sheet (DAG, w.r. paths, min. cost.)

Jun 2024, 3

1) rev. Dijkstra, from all to 7.



3) polym. time alg., DG, costs, vertex 5, shortest cycle of $<$ costs (see ref.).
 $\Rightarrow \Rightarrow$ min. length.

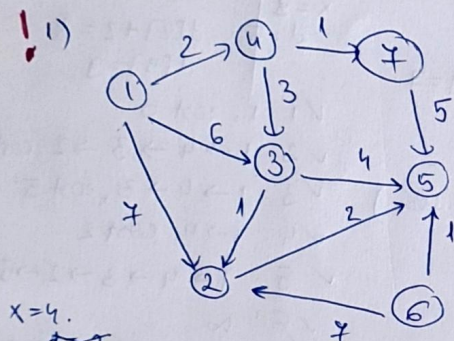
Jun 2018, 9

1) see Jun 2018, 5.

2) see Jun 2018, 5.

3) see ref. sheet.

Jun 2019, 1



$x=4$.
 ~~$d=2$~~
 $d=3$: $d[4]+3 = 2+3=5$
 $d[3]=6$
 $d=4$: $d[4]+1 = 2+1=3$
 $x=3$
 $d=2$: $d[3]+1 = 5+1=6$
 $d[2]=7$
 $d=5$: $d[3]+4 = 5+4=9$
 $d[5]=8$
 $x=2$
 $d=5$: $d[2]=6+2=8$
 $d[5]=8$

x	2	d	prev
		1 2 3 4 5 6 7	1 2 3 4 5 6 7
		0 ∞ ∞ ∞ ∞ ∞ ∞	- - - - -
1	4 3, 2	0 4 6 2 ∞ ∞ ∞	- 1 1 1 - - -
4	3, 2, 7	0 7 5 2 ∞ ∞ 3	- 1 4 1 - - 4
7	2, 2, 5	0 7 5 2 10 ∞ 3	- 1 4 1 7 - 4
3	2, 5	0 6 5 2 8 ∞ 3	- 3 4 1 3 - 4
2	5	0 6 5 2 8 ∞ 3	- 3 4 1 2 - 4
5			

$\Rightarrow$ 1: ~~1 $\rightarrow$ 4 $\rightarrow$ 3 $\rightarrow$ 2~~

$\Rightarrow$ 1: ~~1 $\rightarrow$ 4 $\rightarrow$ 7~~, cost 3.

~~2: 2 $\rightarrow$ 3~~

~~2: 2 $\rightarrow$~~

~~3: 3 $\rightarrow$ 4 $\rightarrow$ 7~~

$\Rightarrow$ 1: ~~1 $\rightarrow$ 4 $\rightarrow$ 7~~, cost 0.

2: 1 $\rightarrow$ 4 $\rightarrow$ 3 $\rightarrow$ 2, cost $\checkmark$

3: 1 $\rightarrow$ 4 $\rightarrow$ 3, cost 5.

4: 1 $\rightarrow$ 4, cost 2.

5: 1 $\rightarrow$ 4 $\rightarrow$ ~~3~~ $\rightarrow$ 5, cost 8

6: ∞

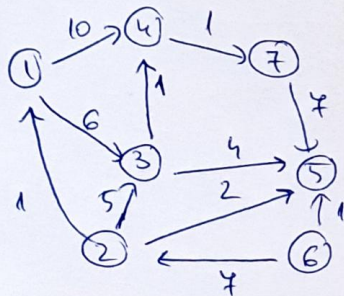
7: 1 $\rightarrow$ 4 $\rightarrow$ 7, cost 3

2) see 15) from „Patrimoniul...“

3) see ref. sheet (DAG, no $\neq$ paths, min. cost.)

Jun 2024, 4

1) rev. Dijkstra, from all to 7.



2) see 16) from „Patrimoniul...“

3) see ref. sheet (DAG, $\neq$ paths, max. len.)